Air Power

Helicopter Briefing

Alan Watson Forster

31 May 2006

INTRODUCTION

This document contains information on helicopters for J.D. Webster's *Air Power* game system. Specifically, it includes the Helicopter Data Table and has notes on each helicopter type, including usual armament.

CONTENTS

HELICOPTERS

The Helicopter Data Table contains basic information on many interesting helicopters. Its columns are:

– The helicopter type and name. The NATO reporting name is given in parentheses.

– Whether it has high speed/agility (HSA).

– Its visibility, size, vulnerability, and blind arc.

– Its characteristics when treated as a ground unit: its soft defense strength (before the dash) and maximum sighting range (after the dash).

– For gun-armed helicopters: the gun type, arc, ammunition, hit rolls, and damage ratings. Door guns are indicated by LS (left side), RSD (right side), and R (rear). The table does not include guns mounted in pods on weapon stations; these are discussed in the notes on each helicopter.

– Its bomb system. B bomb systems have a −1 modifier and C bomb systems have a −2 modifier.

– Its ECM capability: its DDS type, IRM suppression modifier, RWR type, and DJM type. Some helicopters have multiple DDS systems. The IRM suppression modifier is applied to IRM and IR SAM attacks.

– Its technology: whether it has VAS, TV/IR optics, a laser spot tracker, a laser designator, a helmet-mounted sight (HMS), a mast-mounted sight (MMS), or mast-mounted radar (MMR).

GENERIC HELICOPTERS

While the Helicopter Data Table gives data for many specific helicopter types, some will inevitably be missing. To guide in determining data values for other helicopters, the Table gives some typical values for generic helicopters. Examining the data values of other similar helicopters can also provide useful guidance.

The Table gives typical data values for generic scout/utility, medium-lift, and heavy-lift helicopters. If a helicopter has a rear ramp, it is assumed that the load master or door gunner is the searching out of it and so the helicopter has no blind arcs.

The Table gives typical data values for infrared countermeasures from the 1970s (exhaust management and jammer), the 1980s (exhaust management, jammer, and some flares), and the 1990s (exhaust management, jammer, and many flares).

The Table gives typical data values for door guns, specifically a single 7.62 mm LMG with 200 rounds, a .50 cal HMG with 200 rounds, and a 7.62 mm minigun with 1000 rounds.

BRITISH HELICOPTERS

Westland Wessex: The Westland Wessex is a medium anti-submarine, transport, and utility helicopter. It is a licensed version of the Sikorsky S-58 with the radial piston engine replaced by one or two tuboshaft engines.

The HAS.1 and HAS.3 versions are anti-submarine versions. They can be equipped with either a dipping sonar or with torpedos or depth charges, so tend to operate in hunter-killer pairs. The Wessex 31A/B are similar to the HAS.1/3. The HC.2 and HU.5 are transport versions.

The HAS.1/HAS.3/HU.5 served in the RN from 1961 until about 1990. The HC.2 served in the RAF from 1962 to 2022. The HAS.1 and HC2 served in the Indonesia-Malaysia Confrontation. The HAS.3 and HU.5 saw combat in the South Atlantic War.

The Wessex 31A/B served in the RAN in the anti-submarine role 1963 to 1980 and then as a transport and utility helicopter until 1989.

Westland Scout: The Westland Scout AH.1 is a scout, utility, and light attack helicopter developed for the British Army. The similar Wasp was simultaneously developed for the Royal Navy.

It can be armed with a 7.62 mm GPMG door gun, either on the left of the right, with 200 rounds and four SS.11 RGs. Alternatively, it can carry two fixed 7.62 mm GPMG guns, again each with 200 rounds.

The Scout served with the British Army from 1963 to about 1994 and with the Royal Marines from 1971 to 1982. It saw combat in the Indonesia-Malaysia Confrontation, the Aden Emergency, Oman, Northern Ireland, and the South Atlantic War.

Westland Wasp: The Westland Wasp FAS.1 is a anti-submarine and utility helicopter developed for the Royal Navy. The similar Scout was simultaneously developed for the British Army.

For ground-attack and anti-ship missions, it can be armed with a 7.62 mm GPMG door gun, either on the left of the right, with 200 rounds and four SS.11 or two AS.12 RGs. For anti-submarine missions, it can carry two Mk. 44 torpedos, one Mk. 46 torpedo, two Mk. 11 depth charges, or one WE.177 nuclear depth charge.

The Wasp served in the Royal Navy from 1964 to 1988 and saw combat in the South Atlantic War.

Also served with the Royal Malaysian Navy from 1988 to 1998, RNZN from 1966 to 1998, Royal Netherlands Navy from 1966 to 1981, the Indonesian Navy from about 1980 to 1998, the Brazilian Navy from 1966, and the South African Navy from 1963 to 1990.

Westland Sea King: The Westland Sea King is a anti-submarine and medium-lift transport helicopter. It is a licensed adaptation of the Sikorsky S-61, with many British components including the engines and the flight control system.

The HAS.1/2/5/6 differ principally in their anti-submarine systems evolved with time. The HC.4 is a specially adapted version for Commando assault. The HU.5 is an HAS.5 converted to a pure transport and utility helicopter by removing the anti-submarine equipment. The AEW.2/5 and ASaC.7 are airborne early warning and control version, with the Shearwater radar carried in a pod on the right side of the fuselage.

All can be equipped with a pintle-mounted gun in the door on the right side of the fuselage, although this is probably not likely in the AEW versions. In RN service, the 7.62 mm GPMG is used. In service with the German Navy, a .50 cal M2 machine gun is used.

The Sea King HAS served in the RN from 1969 until 2019 and the HC.4 from 1990 to 2016. It saw combat in the South Atlantic War, the Gulf War, the Balkans, during the invasion and occupation of Iraq, and in Afghanistan. Anti-submarine versions were also used by the RAN and the navies of Belgium, Germany, India, Egypt, Norway, Pakistan. Transport versions are used by Egypt, Qatar, and Ukraine.

Westland Lynx: The Westland Lynx is an attack, anti-submarine, anti-ship, and utility helicopter. It was developed by Westland and produced in collaboration with Sud Aviation (which later became part of Aérospatiale). An adapted Lynx holds the world speed record for helicopters.

There are many attack (AH), anti-submarine (HAS), and maritime attack (HMA) versions. The attack versions are armed with four TOW RGs. The naval versions can be armed with torpedos, depth charges, four Sea Skua RGs (in the RN) or four AS-12 missiles (in the French Navy). The naval versions also incorporate a search radar. All can use a door-mounted machine gun, typically a 7.62 mm gun in earlier versions but a .50 cal in the later AH.9A variant.

Important upgrades in the 1990s were the addition of TV/IR Optics to the naval versions and countermeasures in the attack versions.

The Lynx AH entered service in the British Army in 1979. After the introduction of the Apache in 2024, the Lynx was relegated to a utility role and finally retired in 2018. The attack version saw combat in the Gulf War, the invasion and occupation of Iraq, and Afghanistan.

The Lynx HAS/HMA entered service with the Royal Navy in 1981 and was retired in 2017. It saw combat in the South Atlantic War (on both sides). Naval versions also served with the French Navy and many other navies.

FRENCH HELICOPTERS

Sud Aviation Alouette III: The Sud Aviation (later Aérospatiale) Alouette III is a scout, utility, and light attack helicopter. It can carry a pilot and seven passengers. It can be armed with twin 0.303-in Browning machine guns, a single 7.62 mm MAG machine gun, or a 20 mm Matra MG 151 cannon firing out of the left door. Alternatively, it can be armed with 7 × 68 mm RPs, four SS.11 RGs, two AS.12 RGs, or two torpedos.

The Alouette III entered service in France in 1960 and was widely exported. It served in the Rhodesian and South African Bush Wars, where versions with LMG door guns were known as G-Cars and those with 20 mm door guns as K-Cars.

Aérospatiale Puma: The Aérospatiale SA 330 Puma is a medium transport helicopter. It was originally developed by Sud Aviation which later became part of Aérospatiale. It was introduced in 1968 and was widely adapted in the 1970s.

Many versions have improved infrared countermeasures. All offer the ability to mount guns on both doors, although typically only one is mounted.

The variants in RAF service were the HC.1 (which entered service in 1971) and the HC.2 (which entered service in 2012). They saw combat in the Gulf War, the Iraq War, and the Afghanistan War. They were retired in 2025. The Puma also served in the Argentinian forces in the South Atlantic War.

Aérospatiale Gazelle: The Aérospatiale Gazelle is a scout and light attack helicopter. It was originally developed by Sud Aviation which later became part of Aérospatiale as a successor to the Alouette III.

It can be armed with a fixed 20 mm M621 gun, four HOT RGs, or two or four Mistral IRMs. The latest *Viviane* versions in French service have TV/IR optics.

Various versions of Gazelle served in the French Army from 1973 in scout, anti-tank, and anti-helicopter roles. In combat roles, it is being replaced by the Eurocopter Tiger, but still serves in utility roles. The 20 mm, HOT, and dual Mistral versions saw combat in the Gulf War. The Gazelle has also seen combat in many French interventions in former African colonies.

The Gazelle AH.1 served in the British Army from 1973 to 2023 in the scout and utility roles alongside the anti-tank Lynx AH.1. It saw combat in the South Atlantic War, the Gulf War, the Iraq War, and the Afghanistan War.

The Gazelle is widely used by the armed forces of other countries, especially former French colonies.

SOVIET/RUSSIAN HELICOPTERS

Mil Mi-8: The Mil Mi-8 (Hip) is a Soviet/Russian transport and attack helicopters.

The Mi-8T (Hip-C) is a transport version capable of carrying 24 passengers or 4,000 kg of cargo. It is crewed by a pilot, copilot, and flight engineer. Two external weapon racks allow it to be armed with four UB-16-57 16 × 55 mm RPs. A single 7.62 mm PK machine gun can be mounted in place of the central lower window and operated by the flight engineer, two more can be mounted on the weapon racks, and up to three more can be mounted on the side and rear doors.

The Mi-8TV (Hip-E) version is a dedicated attack helicopter; when carrying a complete load of weapons it cannot carry a useful load of passengers or cargo. The forward firing PK machine gun is replaced by a 12.7 mm VK-4 heavy machine gun, the stores racks are extended to allow carriage of six UB-16-57 16 × 55 mm RPs, and four 9M17P Falanga-P (AT-2C Swatter-C) missiles can be carried above the racks.

The Mi-8TVK (Hip-F) is the export version of the Mi-8TV and substitutes six 9M914M Malyutka (AT-3 Sagger) missiles for the Falangas.

As a result of experience in the Soviet-Afghan War, many Mi-8 helicopters were fitted with improved countermeasures, including an exhaust cooling system, an infrared jammer, flares, and a radar-warning system.

The Mi-8T entered service in the Soviet Union in 1967 and the Mi-8TV in 1974. They saw combat in the Soviet-Afghan War. After the breakup of the Soviet Union, they saw combat in the many wars between Russia and other successor states. The Mi-8 has been widely exported, both to allies of the Soviet Union or Russia and to many neutral countries.

Mil Mi-24: The Mil Mi-24 is a Soviet/Russian attack and assault helicopter. Uniquely, it can carry eight passengers in addition to its air-to-ground weapons.

The first Mi-24 (Hind-A) version has a flight crew of three, pilot, a navigator, and a gunner. While it is fast and very well protected by armor, it is not very agile. It can carry four 9M17 Falanga-M (AT-2 Swatter-2) RGs on the wing-tip stations and four UB-32 32 × 55 mm or B-8V20A 20 × 80 mm RPs on its under-wing stations. It also has a single-barrelled 12.7 mm A-12.7 gun in a turret under the nose. The

The Mi-24D (Hind-D) has a reduced crew of a pilot and gunner and accommodates them in a completely redesigned cockpit. The 12.7 mm A-12.7 gun was replaced by a multi-barrel 12.7 mm YakB-12.7 gun and the missiles upgraded to the much more effective 9M17P Falanga-P (AT-2C Swatter). The Mi-24V (Hind-E) uses the 9M114 Kokon (AT-6 Spiral) missile, but is otherwise very similar.

The Mi-24P (Hind-F) replaces the 12.7 mm gun with a fixed two-barrelled 30 mm GShK-30-2K cannon, but is otherwise similar to the Mi-24V.

As a result of experience in the Soviet-Afghan War, many Mi-

24D/V helicopters were fitted with improved countermeasures, including an exhaust cooling system, flares, and a radar warning system. These were fitted as standard on the Mi-24P.

The Mi-24 and Mi-24D entered service with Soviet forces in 1973, the Mi-24V in 1976, and the Mi-24P in 1981. In Soviet service, they saw combat in the Soviet-Afghan War. They also served in the armed forces of the Soviet Union's successor states and in many allies of the Soviet Union, and saw combat notably in the Iran-Iraq War, the Gulf War, and the Invasion of Ukraine.

Mil Mi-28: The Mil Mi-28 is an attack helicopter. It was originally designed in the 1980s, following the collapse of the Soviet Union its finally entered service in 2010. The Mi-28 is a significant departure from the earlier Mi-24 and in many ways resembles the AH-64D Apache, with two crew members, heavy armor, a mast-mounted radar, and a 30 mm cannon.

The Mi-28N (Havoc-B) version has four weapon stations, each of which can carry eight 9M120 Ataka (AT-9 Spiral-2) missiles, a B-8V20A 20 × 80 mm RP, or a 23 mm UPK-23-250 GP. The Mi-28NE is a similar export version.

The Mi-28N entered service with the Russian armed forces in 2010 and has seen combat in the Invasion of Ukraine. The Mi-28NE serves in the armed forces of Iraq and Algeria.

US HELICOPTERS

Bell UH-1 Iroquois: The Bell UH-1 Iroquois is a medium transport and attack helicopter. It was introduced in 1959 with the US Army as the HU-1 and subsequently was almost universally known by its nickname "Huey" rather than its official name "Iroquois". In 1962, it was redesignated as UH-1. The D and H versions had stretched cabins to accommodate more passengers.

It is often equipped with two 7.62 mm M60 door guns. During the Vietnam War, some B and C versions were also used as gunships or "airborne rocket artillery".

The principal gunship variants use the M6/M16 armament subsystem with four 7.62 mm M60 machine guns or two 7.62 mm M134 miniguns on two forward-facing articulated mounts either side of the fuselage. The gunships retain their door guns and the M16/M21 variants could also carry two fixed M158 7 × 70 mm RPs, one on each side of the fuselage.

The principal rocket variant uses two fixed M3 24 × 70 mm RPs, one on each side of the fuselage. Again, it retains the door guns.

The M11/M22 armament subsystems allow the UH-1 to be armed with six the AGM-22A/AGM-22B ATGMs (licensed versions of the French SS.11 ATGM) on racks on either side of the fuselage. The experimental XM26 armament subsystem likewise allows it to use six TOW missiles.

It saw extensive service in the US Army, USMC, and USAF from 1961 until being finally withdrawn in the 21st century, and combat in almost all the wars in which the US was involved during that period. Notably, versions armed with the AGM-22B saw service in South Vietnam in 1966 to 1967 and 1972 and those with TOW in 1972. It also served with many US allies, especially the ARVN in the Vietnam War.

Bell AH-1 Cobra: The Bell AH-1 Cobra is an attack helicopter originally derived from the UH-1 Iroquois utility helicopter.

The first AH-1G version is an interim attack helicopter designed to escort UH-1s in the Vietnam War. It uses the drive train of the UH-1 with a new, thin, armored cockpit. It is armed with a turret mounting two 7.62 mm M134 miniguns (or one minigun and one rocket launcher). Its stub wings have four weapon stations for two 19 × 70 mm and two 7 × 70 mm RPs or alternatively two M18 minigun pods in place of the larger RPs. The 1969 M35 variant has a 20 mm M195 Vulcan cannon on the left inner pylon and provides 950 rounds of ammunition. The 1972 upgrade variant has an AN/ALQ-144 IRCM unit and modified exhaust system to defend against the SA-7 IRM.

After the Vietnam War, the US Army refocused on a possible war in Europe, and adapted AH-1s to carry the TOW missile. These were the AH-1Q/S/P/E/F versions, each of which could carry eight TOW RGs. There were many incremental improvements in this series, but some notable differences were that the S/P could not carry RPs and the E adopted the 20 mm M197 cannon from the AH-1T Sea Cobra.

The AH-1G entered service with the US Army in 1967 and the AH-1Q/S/P/E/F in the late 1970s. The AH-1 was retired by the US Army in 1999.

Bell AH-1 Sea Cobra, Super Cobra, and Viper: The Bell AH-1 Sea Cobra is a two-engined version of the AH-1 Cobra developed for the USMC.

The AH-1J is armed with a turret mounting the 20 mm M197 cannon instead of two miniguns, but its armament is otherwise largely similar to the AH-1G. Its stub wings have four weapon stations for two 19 × 70 mm and two 7 × 70 mm RPs. The 1972 upgrade variant has an AN/ALQ-144 IRCM unit and modified exhaust system to defend against the SA-7 IRM.

The AH-1J was developed into the AH-1T, which has the capacity to carry two launchers each with four TOW RGs in place of the larger RPs.

The AH-1W Super Cobra is based on the AH-1T, but has significantly updated systems. In addition to the usual RPs, it can also carry two launchers each for four AGM-114 Hellfire RGs (in place of the TOW missile), two AIM-9L IRM (one on each of the outer stations), and AGM-122 ARM (one on each of the outer stations).

The AH-1Z Viper is another iteration, with improved systems and two new wing-tip stations that can carry either AIM-9L IRMs or (on the right side) the Longbow attack radar. It can also carry up to sixteen AGM-114 Hellfire RGs (one launcher on each wing station each with four missiles) or eight AGM-179 JAGM RG missiles (one launcher on each inner wing station each with four missiles).

The AH-1J entered service in 1970, the AH-1T in the late 1970s, the AH-1W in 1986, and the AH-1Z in 2011.

Sikorsky CH-3/HH-3: The Sikorsky CH-3/HH-3 is a medium transport and SAR helicopter. It was developed from the SH-3 Sea King, and one significant change was the addition of a rear ramp. It was known as the "Jolly Green Giant".

The CH-3C is the initial version and entered service in the USAF in 1964. The CH-3E has more powerful motors. The HH-3E is a conversion of the CH-3E with protective armor, self-sealing fuel tanks, an inflight refueling probe, a high-speed hoist, and other equipment for their SAR role.

The HH-3E is armed with three 7.62 mm M60 door guns, firing to the left, right, and rear. These can also be fitted to the CH-3E.

The CH-3/HH-3 served in the USAF for special operations and SAR, notably during the Vietnam War. The HH-3 was later replaced by the HH-53.

Hughes OH-6 Cayuse: The Hughes OH-6 is a scout and light attack helicopter. Although formally named the "Cayuse", it was more commonly known as the “Loach,” with this nickname being derived from the *Light Observation Helicopter* program under which it was developed.It is small, maneuverable, and robust.

While normally unarmed, the OH-6A can be fitted with the M27 armament subsystem that mounts a 7.62 mm M134 minigun on the right side of the fuselage and provides it with 2000 rounds of ammunition. The mount can elevate and depress, but not traverse.

The OH-6A version served in large numbers with the US Army in the Vietnam War, entering service in 1966 and serving until being replaced by the OH-58 towards the end of the war.

Boeing CH-47 Chinook: The Boeing CH-47 is a twin-rotor, heavy-lift transport helicopter.

Although the CH-47 has evolved and adapted more modern technologies over its long life, the CH-47A/B/C/D/F/J and HC.1/2/4/5/6 versions have similar capabilities in game terms and are not distinguished in the Helicopter Data Table. Later models, often upgraded in service, have improved infrared countermeasures. The M24 and M41 armament subsystems mount 7.62 mm M60 or M240 machine guns on the side (M24) and rear doors (M41).

The ACH-47 “Guns-a-Go-Go” gunship has additional armor and the M34 armament subsystem providing two fixed 20 mm M24 cannon, and provision for two M159 19 × 70 mm RPs. It also had .50 cal M2 door guns provided by the M32 and M33 armament subsystems.

The MH-47 versions are adapted for special-forces operations. The MH-47D has air-to-air refueling, forward-looking infrared, and the side M60s are replaced by M134 miniguns, the HH-47E adds terrain-following radar, and the MH-47G is armed with two 7.62 mm M134 miniguns and two 7.62 mm M240 machine guns on the side doors.

The CH-47 entered service with the US Army in 1966 and served in the Vietnam War and many conflicts since then. It also serves or has served with many US allies, including the RAF. Four ACH-47s were tested by the USAF in Vietnam in 1966. The MH-47D entered service in 1984, the MH-47E in 1991 and the MH-47F in 2006.

Sikorsky CH-53 Sea Stallion and HH-53: The Sikorsky CH-53/HH-53 is a heavy transport and SAR helicopter. It was developed from the CH-3 and the CH-54. The CH-53 version is known as the Sea Stallion, and the HH-53 versions are not officially named but are known informally as the “Super Jolly Green Giant” or “Super Jolly”.

The CH-53 was developed as a heavy-lift transport helicopter for the USMC. The A and D versions served with the USMC, the D version with the Israeli Air Force, and the G version with the German Army.

The USAF versions are the HH-53B, HH-53C, and CH-53C. The HH-53B CSAR version, delivered in 1968, has a refueling probe, drop tanks, armor, a rescue hoist, and three 7.62 mm miniguns on the left, right, and rear doors. The HH-53C version, also delivered in 1968, has smaller external fuel tanks, but more armor and better communications equipment. The CH-53C version was similar to the HH-53C, including armor and door guns, but lacks the refueling probe.

The HH-53B and HH-53C served in the USAF in CSAR roles, replacing the HH-3E, and saw combat in the Vietnam War.

Bell OH-58 Kiowa: The Bell OH-58 Kiowa is a scout and light attack helicopter. It initially lost the *Light Observation Helicopter* competition, but was later ordered by the US Army when the rising cost of the Hughes OH-6 caused bidding to be repeated.

The OH-58A is the initial version. Like the OH-6, it can be fitted with the M27 armament subsystem that mounts a 7.62 mm M134 minigun on the right side of the fuselage and provides it with 2000 rounds of ammunition. The mount can elevate and depress, but not traverse.

The OH-58C version is a conversion of the OH-58A and adds an IR suppression system and RWR. It may be armed with the M27 subsystem or two AIM-92 Stinger IRMs.

The OH-58D version in turn is a modification of the OH-58C and adds a more powerful engine, a four-bladed main rotor, and a mast-mounted sight. The *Armed* and *Kiowa Warrior* variants have two mounts on the fuselage sides each of which can carry two AGM-114 Hellfire RGs, two AIM-92 Stinger IRMs, or one M260 7 × 70 mm RP. The left pylon can alternatively carry an M296 0.50 cal GP with 500 rounds. The *Kiowa Warrior* variant also has improved defenses against IRMs.

The OH-58A served with the US Army from 1969 and saw combat in the Vietnam War. The C and D versions entered service in 1976 and 1986, and saw combat in both Gulf Wars. The *Armed* was developed in 1987, and these variants saw combat in the Persian Gulf protecting oil tankers from attacks by Iranian forces. There are suggestions that the aircraft flew in pairs, one armed with two Hellfire missiles and a .50 cal machine gun and the other with two Stingers and one rocket pod. The *Kiowa Warrior* variant entered service in 1989 and saw combat in the Second Gulf War, in the Occupation of Iraq, and in Afghanistan.

Hughes 500 Defender: The Hughes 500 Defender is a light attack helicopter. It was developed in the 1970s from the civilian Hughes 500 which in turn was derived from the OH-6.

The Defender has two weapon stations on the fuselage sides each of which can carry two TOW RGs, one M27 GP with a 7.62 mm M134 minigun with 2000 rounds, one FN HMP400 0.50 cal GP with 400 rounds, one 7×70 mm RP, or two AIM-92 Stinger IRMs.

The Defender served with many armed forces and saw combat with Israeli Air Force against Syrian forces in the late 1970s and 1980s.

Hughes AH-64 Apache: The Hughes AH-64 Apache is an attack helicopter designed specifically for the anti-tank role. Hughes Helicopters was later taken over by McDonnell Douglas which in turn was taken over by Boeing, so the Apache is now produced and developed by Boeing.

The initial AH-64A version is armed with a 30 mm M230 chain gun in a swivel mount under the nose and 1200 rounds. Its stub wings have four weapon stations, each of which can carry four AGM-114 Hellfire RGs, a M261 19 × 70 mm RPs, or a dual rack for AIM-92 Stinger IRMs. It entered service in the US Army in 1984 and was declared operational in 1986. It served in the First Gulf War and other US operations in the 1990s, and was finally retired by the US Army in 2012.

The AH-64D/E versions have the Longbow mast-mounted radar that allows targets to be identified and engaged from behind defilade. The radar can be removed in low-threat environments. The AH-64D entered service with the US Army in 1997 and saw combat in the Second Gulf War, the Occupation of Iraq, and Afghanistan.

Helicopter Data Table

	Aircraft Data						Gun						IRCM & ECM				Technology							
Type	HSA	Visibility	Size	Vulnerability	Blind Arc	Ground Unit Data	Type	Arc	Ammunition	Hit	AtA/AtG Damage	Bomb System	DDS	IRM Suppression	RWR	DJM	VAS	TV/IR Optics	Laser Spot Tracker	Laser Designator	HMS	MMS	MMR	TFR
Generic Helicopters																								
Scout or Utility Helicopter	—	5	+0	−2	30°–	4-8	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
Medium-Lift Helicopter	—	6	+0	+0	60°–	4-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
Heavy-Lift Helicopter	—	7	+0	+0	60°–	3-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
Rear ramp					—																			
IRCM: exhaust mitigation														+1										
IRCM: jammer														+1										
IRCM: early DDS													A × 1											
IRCM: later DDS													A × 2											
Door gun: LMG with 200 rounds							7.62 mm LMG	LS, RS, or R	10	1/1/–	1/1**													
Door gun: HMG with 200 rounds							.50 cal HMG	LS, RS, or R	12	1/1/–	2/1**													
Door gun: Minigun with 3000 rounds							7.62 mm Minigun	LS, RS, or R	30	1/1/–	2/2**													
British Helicopters																								
Wessex HAS.1/HC.2/HAS.3/HU.5	—	6	+0	+0	60°–	4-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
Scout AH.1	—	4	+1	−2	30°–	3-6	—	—	—	—	—	—	—	+0	—	—	Y	—	—	—	—	—	—	—
Scout AH.1 (Door GPMG)							7.62 mm GPMG	LS or RS	10	1/1/–	1/1**													
Scout AH.1 (Fixed GPMGs)							7.62 mm GPMG × 2	Fixed	20	2/1/–	1/1**													
Wasp HAS.1	—	4	+1	−2	30°–	3-6	—	—	—	—	—	—	—	+0	—	—	Y	—	—	—	—	—	—	—
Wasp HAS.1 (Door GPMG)							7.62 mm GPMG	LS or RS	10	1/1/–	1/1**													
Sea King HAS/HC/HU/AEW	—	6	+0	+0	60°–	4-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
Sea King HAS/HC/HU (Door GPMG)							7.62 mm GPMG	RS	10	1/1/–	1/1**													
Sea King HAS (Door M2)							.50 cal M2	RS	12	1/1/–	2/1**													
Sea King HU (1990s IRCM)													A × 2	+0	—	—								
Lynx AH/HAS/HMA	Y	5	+0	+0	30°–	4-12	—	—	—	—	—	—	—	+0	—	—	Y	—	—	—	—	—	—	—
Lynx AH/HAS/HMA (Door GPMG)							7.62 mm GPMG	RS	10	1/1/–	1/1**													
Lynx AH.9A (Door M2)							.50 cal M2	RS	12	1/1/–	2/1**													
Lynx AH (1990s IRCM)													A × 2	+0	—	—								
Lynx HAS/HMA (1990s FLIR)																	Y	Y	—	—	—	—	—	—

Helicopter Data Table

	Aircraft Data						Gun						IRCM & ECM				Technology							
Type	HSA	Visibility	Size	Vulnerability	Blind Arc	Ground Unit Data	Type	Arc	Ammunition	Hit	AtA/AtG Damage	Bomb System	DDS	IRM Suppression	RWR	DJM	VAS	TV/IR Optics	Laser Spot Tracker	Laser Designator	HMS	MMS	MMR	TFR
French Helicopters																								
Alouette III	—	4	+1	−1	30°–	4-8	—	—	—	—	—	M	—	+0	—	—	—	—	—	—	—	—	—	—
Alouette III (Browning door gun)							0.303 in Browning × 2	LS	10	1/1/—	1/1													
Alouette III (MAG door gun)							7.62 mm MAG × 2	LS	10	1/1/—	1/1													
Alouette III (MG 151 door gun)							20 mm MG 151	LS	10	1/1/1	3/4*													
Alouette III (ATGM sight)																	Y	—	—	—	—	—	—	—
Puma	—	6	+0	+0	60°–	4-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
Puma (1980s IRCM)													A × 1	+0	—	—								
Puma (1990s IRCM)													A × 2	+0	—	—								
Puma (Door LMG)							7.62 mm LMG	LS or RS	10	1/1/–	1/1**													
Gazelle	Y	4	+1	−1	30°–	4-8	—	—	—	—	—	—	—	+1	—	—	Y	—	—	—	—	—	—	—
Gazelle (1990s IRCM/ECM)													A × 1	+1	A	—								
Gazelle (M621)							20 mm M621	Fixed	15	3/2/1	3/4*													
Gazelle (Viviane)																	Y	Y	—	—	—	—	—	—

Helicopter Data Table

Type	Aircraft Data						Gun						IRCM & ECM				Technology							
	HSA	Visibility	Size	Vulnerability	Blind Arc	Ground Unit Data	Type	Arc	Ammunition	Hit	AtA/AtG Damage	Bomb System	DDS	IRM Suppression	RWR	DJM	VAS	TV/IR Optics	Laser Spot Tracker	Laser Designator	HMS	MMS	MMR	TFR
Soviet and Russian Helicopters																								
Mi-8T Hip-C	—	6	+0	+0	60°–	4-12	—	—	—	—	—	M	—	+0	—	—	—	—	—	—	—	—	—	—
Mi-8T Hip-C (Forward PK)							7.62 mm PK	180°+	10	1/1/–	1/1**													
Mi-8T Hip-C (Forward and Rack PKs)							7.62 mm PK × 3	Fixed	10	2/1/–	2/2**													
Mi-8T Hip-C (Door PKs)							7.62 mm PK × 3	RS/LS/R	10	1/1/–	1/1**													
Mi-8TV/TBK Hip-E/F	—	6	+0	+0	60°–	4-12	12.7 mm VK-4	180°+	10	1/1/–	1/2**	M	—	+0	—	—	Y	—	—	—	—	—	—	—
Mi-8T/TV/TBK Hip-C/E/F (1980s IRCM/ECM)													A × 2	+2	B	—								
Mi-24 Hind-A	—	6	+0	+2	60°–	6-12	12.7 mm A-12.7	150°+	10	4/2/–	1/2**	B	—	—	—	—	Y	—	—	—	—	—	—	—
Mi-24D Hind-D	—	6	+0	+2	60°–	6-12	12.7 mm YakB-12.7	150°+	10	5/3/–	2/4**	B	—	—	—	—	Y	Y	—	—	—	—	—	—
Mi-24D Hind-D (1980s IRCM/ECM)													A × 2	+2	B	—								
Mi-24V Hind-E	—	6	+0	+2	60°–	6-12	12.7 mm YakB-12.7	150°+	10	5/3/–	2/4**	B	—	—	—	—	Y	Y	—	—	—	—	—	—
Mi-24V Hind-E (1980s IRCM/ECM)													A × 2	+2	B	—								
Mi-24P Hind-F	—	6	+0	+2	60°–	6-12	30 mm GShK-30-2K	Fixed	10	3/2/1	6/8	B	A × 2	+2	B	—	Y	Y	—	—	—	—	—	—
Mi-28N/NE Havoc-B	Y	5	+0	+3	30°–	6-12	30 mm 2A42	150°+	10	5/3/2	6/6	C	A × 2	+2	B	—	Y	Y	Y	—	Y	—	Y	—

Helicopter Data Table

Type	Aircraft Data						Gun						IRCM & ECM				Technology							
	HSA	Visibility	Size	Vulnerability	Blind Arc	Ground Unit Data	Type	Arc	Ammunition	Hit	AtA/AtG Damage	Bomb System	DDS	IRM Suppression	RWR	DJM	VAS	TV/IR Optics	Laser Spot Tracker	Laser Designator	HMS	MMS	MMR	TFR
US Helicopters																								
UH-1B/C/D/H Iroquois	—	5	+0	+0	30°–	4-12	7.62 mm M60 × 2	LS/RS	10	1/1/–	1/1**	—	—	+0	—	—	—	—	—	—	—	—	—	—
UH-1B/C Iroquois (M3)							—	—	—	—	—	B												
UH-1B/C Iroquois (M6/M16)							7.62 mm M60 × 4	180°+	35	3/2/–	2/2**	B												
UH-1B/C Iroquois (M21)							7.62 mm M134 × 2	180°+	30	3/2/–	3/3**	B												
UH-1B/C Iroquois (M11/M22/XM26)												B					Y	—	—	—	—	—	—	—
AH-1G Cobra	Y	5	+0	+1	30°–	5-8	7.62 mm M134 × 2	120°+	20	5/3/–	3/3**	B	—	+0	—	—	—	—	—	—	—	—	—	—
AH-1G Cobra (M35)							20 mm M195	Fixed	5	3/2/–	6/8*													
AH-1G Cobra (1972 Upgrade)													—	+2	—	—								
AH-1Q/S Cobra	Y	5	+0	+1	30°–	5-8	7.62 mm M134 × 2	120°+	20	5/3/–	3/3**	B	—	+2	—	—	Y	—	—	—	—	—	—	—
AH-1P Cobra	Y	5	+0	+1	30°–	5-8	7.62 mm M134 × 2	120°+	20	5/3/–	3/3**	B	—	+2	A	—	Y	—	—	—	—	—	—	—
AH-1E/F Cobra	Y	5	+0	+1	30°–	5-8	20 mm M197	150°+	30	5/3/2	3/4*	B	—	+2	A	—	Y	—	—	—	Y	—	—	—
AH-1J Sea Cobra	Y	5	+0	+1	30°–	5-8	20 mm M197	150°+	30	5/3/2	3/4*	B	—	+0	—	—	—	—	—	—	—	—	—	—
AH-1J Sea Cobra (1972 Upgrade)													—	+2	—	—								
AH-1T Sea Cobra	Y	5	+0	+1	30°–	5-8	20 mm M197	150°+	30	5/3/2	3/4*	B	—	+2	—	—	Y	—	—	—	—	—	—	—
AH-1W Super Cobra	Y	5	+0	+1	30°–	5-8	20 mm M197	150°+	30	5/3/2	3/4*	C	A × 2	+2	B	—	Y	Y	—	B	—	—	—	—
AH-1Z Viper	Y	5	+0	+1	30°–	5-8	20 mm M197	150°+	30	5/3/2	3/4*	C	A × 2	+2	B	—	Y	Y	—	B	Y	—	—	—
CH-3C	—	6	+0	+0	—	4-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
CH-3E	—	6	+0	+0	—	4-12	7.62 mm M60 × 3	LS/RS/R	35	1/1/–	1/1**	—	—	+0	—	—	—	—	—	—	—	—	—	—
HH-3E	—	6	+0	+1	—	4-12	7.62 mm M60 × 3	LS/RS/R	35	1/1/–	1/1**	—	—	+0	—	—	—	—	—	—	—	—	—	—
OH-6A Cayuse	—	4	+1	−1	30°–	3-6	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
OH-6A Cayuse (M27)							7.62 mm M134	Fixed	20	3/2/–	2/2**													
CH-47 Chinook	—	7	+0	+0	—	4-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
CH-47 Chinook (1980s IRCMs)													A × 1	+2	—	—								
CH-47 Chinook (1990s IRCMs)													A × 2	+2	—	—								
CH-47 Chinook (M24/M41)							7.62 mm M60/M240 × 3	LS/RS/R	10	1/1/–	1/1**													
ACH-47A Chinook	—	7	+0	+1	—	4-12	20 mm M24 × 2	Fixed	20	3/2/1	4/6*	B	—	+0	—	—	—	—	—	—	—	—	—	—
							.50 cal M2 × 3	LS/RS/R	12	1/1/–	2/1**													
MH-47D Chinook	—	7	+0	+1	—	4-12	7.62 mm M134 × 2	LS/RS	30	1/1/–	2/2**	—	A × 1	+2	—	—	Y	—	—	—	—	—	—	—
							7.62 mm M60	R	35	1/1/–	1/1**													
MH-47E Chinook	—	7	+0	+1	—	4-12	7.62 mm M134 × 2	LS/RS	30	1/1/–	2/2**	—	A × 2	+2	—	—	—	Y	—	—	—	—	—	Y
							7.62 mm M60	R	35	1/1/–	1/1**													
MH-47G Chinook	—	7	+0	+1	—	4-12	7.62 mm M134 × 2	LS/RS	30	1/1/–	2/2**	—	A × 2	+2	—	—	—	Y	—	—	—	—	—	Y
							7.62 mm M240 × 2	LS/RS	35	1/1/–	1/1**													

Helicopter Data Table

	Aircraft Data						Gun						IRCM & ECM				Technology							
Type	HSA	Visibility	Size	Vulnerability	Blind Arc	Ground Unit Data	Type	Arc	Ammunition	Hit	AtA/AtG Damage	Bomb System	DDS	IRM Suppression	RWR	DJM	VAS	TV/IR Optics	Laser Spot Tracker	Laser Designator	HMS	MMS	MMR	TFR
US Helicopters (Continued)																								
CH-53A/D/G Sea Stallion	—	7	+0	+0	—	4-12	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
HH-53B	—	7	+0	+1	—	4-12	7.62 mm GAU-2 × 3	LS/RS/R	8	1/1/–	2/2**	—	—	+0	—	—	—	—	—	—	—	—	—	—
HH-53C/CH-53C	—	7	+0	+2	—	4-12	7.62 mm GAU-2 × 3	LS/RS/R	8	1/1/–	2/2**	—	—	+0	—	—	—	—	—	—	—	—	—	—
OH-58A/B Kiowa	—	4	+1	−2	30°–	3-6	—	—	—	—	—	—	—	+0	—	—	—	—	—	—	—	—	—	—
OH-58C Kiowa	—	4	+1	−2	30°–	3-6	—	—	—	—	—	—	A × 1	+1	A	—	—	—	—	—	—	—	—	—
OH-58D Kiowa	—	4	+1	−2	30°–	3-6	—	—	—	—	—	—	A × 1	+1	A	—	Y	Y	Y	B	—	Y	—	—
OH-58A/B/C/D Kiowa (M27)							7.62 mm M134	Fixed	20	3/2/–	2/2**													
OH-58D Kiowa (Armed)	—	4	+1	−2	30°–	3-6	—	—	—	—	—	B	A × 1	+1	A	—	Y	Y	Y	B	—	Y	—	—
OH-58D Kiowa (Armed M296)							.50 cal M262	Fixed	20	3/2/–	2/1**													
OH-58D Kiowa Warrior	—	4	+1	−2	30°–	3-6	—	—	—	—	—	B	A × 1	+2	A	—	Y	Y	Y	B	—	Y	—	—
OH-58D Kiowa Warrior (M296)							.50 cal M262	Fixed	20	3/2/–	2/1**													
Defender	—	4	+1	−1	30°–	3-6	—	—	—	—	—	B	A × 1	+1	—	—	—	—	—	—	—	—	—	—
Defender (M27)							7.62 mm M134	Fixed	20	3/2/–	2/2**													
Defender (HMP400)							.50 cal M3P	Fixed	16	3/2/–	2/1**													
Defender (TOW sight)																	Y	—	—	—	—	—	—	—
AH-64A Apache	Y	5	+0	+2	30°–	5-8	30 mm M230	120°+	55	5/3/2	6/6	C	A × 2	+3	A	B	Y	Y	Y	B	Y	—	—	—
AH-64D/E Apache	Y	5	+0	+2	30°–	5-8	30 mm M230	120°+	55	5/3/2	6/6	C	A × 2	+3	A	B	Y	Y	Y	B	Y	—	Y	—

HELICOPTER STORES

Helicopter ATGM Table

Type		Year	Guidance	Modifier	Range	Attack Strength			Warhead
						Soft	Hard	Vehicle	
French ATGMs									
SS.11 (AGM-22A)		1956	RCG(B)	−3	1–6	0.5	0.5	3.0	S
SS.11B (AGM-22B)		1962	RCG(B)	−3	1–6	0.5	0.5	3.0	S
AS.12		1960	RCG(B)	−3	4–14	0.5	0.5	3.0	S
HOT		1977	ACG(B)	−4	0–8	0.5	0.5	3.0	S
HOT-2		1985	ACG(B)	−4	0–8	0.5	0.5	3.0	S
HOT-3		1998	ACG(C)	−5	0–8	0.5	0.5	3.0	T
Soviet/Russian ATGMs									
9M11 Fleyta	AT-2 Swatter	1964	RCG(B)	−3	1–5	0.5	0.5	3.0	S
9M14 Malyutka	AT-3 Sagger	1963	RCG(M)	−2	1–6	0.5	0.5	3.0	S
9M14M Malyutka-M	AT-3B Sagger-B	1973	RCG(M)	−2	1–6	0.5	0.5	3.0	S
9M14P Matyutka-P	AT-3C Sagger-C	1969	ACG(B)	−4	1–6	0.5	0.5	3.0	S
9M14-2 Malyutka-2	AT-3D Sagger-D	1992	ACG(B)	−4	1–6	0.5	0.5	3.0	T
9M17M Falanga-M	AT-2B Swatter-B	1968	RCG(B)	−3	1–7	0.5	0.5	3.0	S
9M17P Falanga-P	AT-2C Swatter-C	1969	ACG(B)	−4	1–7	0.5	0.5	3.0	S
9M114 Kokon	AT-6 Spiral	1976	ACG(B)	−4	1–10	0.5	0.5	3.0	S
9M114M1 Kokon	AT-6B Spiral-B		ACG(B)	−4	1–12	0.5	0.5	3.0	S
9M114M2 Kokon	AT-6C Spiral-C		ACG(B)	−4	1–14	0.5	0.5	3.0	S
9M120 Ataka	AT-9 Spiral-2	1985	ACG(C)	−5	1–12	0.5	0.5	3.0	T
9M120M Ataka	AT-9 Spiral-2		ACG(C)	−5	1–14	0.5	0.5	3.0	T
9M120D Ataka	AT-9 Spiral-2		ACG(C)	−5	1–16	0.5	0.5	3.0	T
9M121 Vikhr	AT-16 Scallion	1985	ACG(C)	−5	1–16	0.5	0.5	3.0	T
9M123 Khrizantema	AT-15 Springer	2005	ACG(C)	−5	1–16	0.5	0.5	3.0	T
US ATGMs									
BGM-71A/B/C TOW		1972	ACG(B)	−4	0–7	0.5	0.5	3.0	S
BGM-71D TOW-2A		1983	ACG(B)	−4	0–7	0.5	0.5	3.0	S
BGM-71E TOW-2A		1987	ACG(B)	−4	0–7	0.5	0.5	3.0	S
BGM-71F TOW-2B		1987	ACG(B)	−4	0–9	0.5	0.5	4.0	TA
AGM-114A/B/C Hellfire		1984	LG(C)	−6	1–16	0.5	0.5	3.0	S
AGM-114F/FA Hellfire		1991	LG(C)	−6	1–16	0.5	0.5	3.0	T
AGM-114K Hellfire II		1993	LG(C)	−6	1–22	0.5	0.5	3.0	T
AGM-114L Hellfire Longbow		1995	RS(C)	−6	1–16	0.5	0.5	3.0	T

Guidance: RCG = remote-control guidance, AGC = automatic-control guidance, LG = laser guidance, RS = radar smart guidance, (M) = manual bomb system, (B) = ballistic bomb system, and (C) = computed bomb system. Warhead: S = Singular, T = Tandem, and TA = Top-Attack.

Helicopter RP Table

Type	Rockets	Attack Strength	
		Soft	Hard
French RPs			
Matra 122	7 × 68 mm	2.0	1.0
Soviet/Russian RPs			
UB-16-57	16 × 55 mm	4.0	1.0
UB-32-57	32 × 55 mm	8.0	2.5
B-8V20A	20 × 80 mm	9.0	4.5
US RPs			
M158/M260	7 × 70 mm	2.0	1.0
M159/M200/M261	19 × 70 mm	6.0	3.0
M3	24 × 70 mm	8.0	4.0

Helicopter GP Table

Type	Shots	Hit	Damage ATA/ATG
Soviet/Russian GPs			
23 mm UPK-23-250	2.5	3/2/1	5/4
US GPs			
7.62 mm M18	15	3/2/–	2/2